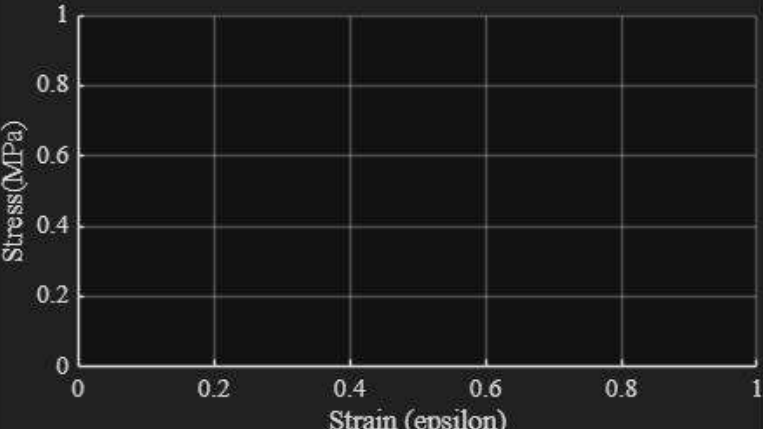


Robot Arm Optimization App Project

Introduction

The app intends to facilitate the process of manufacturing robot arms by applying required stress analysis built in the App.

How it works

Main Parameters	Stress-Strain Table
Croos Section Type Choose type ▼ Material of Arm Choose Material ▼ Radius (cm) 0 Height (cm) 0 Width (cm) 0 Length (cm) 0 Max accelration (rad/sec^2) 0 Load Mass (kg) 0 Calculate	Stress(MPa) Strain (epsilon)  

(Application user interface)

Firstly, the user chooses the cross-section type of arm whether it's circular or rectangular.

Secondly, he chooses from a variety of materials to decide which one the arm will be built by. Also, there's a strain-strain curve for each material that appears when choosing the material to facilitate the decision-making process.

Lastly, the user enters the dimension of the arm that is in the prototype, the max acceleration intended to be operated on, and the mass to be held by the arm.

The display section

Optimized Arm	
Length(cm)	0
Radius(cm)	0
Width(cm)	0
Height(cm)	0
Mass of Arm(kg)	0
Max Stress (MPa)	0

(Display App)

After pressing the calculate button, the final optimized dimensions, mass and the maximum stress of the arm.